When Do Activation-Patching Effects Add Up? Evidence from Controlled CNNs

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Abstract

Purpose. Can the effect of patching several activation channels be predicted from the effects of patching them individually? We investigate when intervention-size dependence reflects the evaluation metric and when it requires non-additive changes in a network's logits. **Methods.** We train small convolutional neural networks initialized with a bank of predefined edge, corner, and ring filters. Retention maintains a first-layer template penalty, whereas release anneals that penalty to zero. We compare the observed effects of joint first-layer patches with reconstructions obtained by summing individual centered-logit changes, across two controlled tasks, 16 architectures, and a pixel-permuted filter-bank control that preserves the Gram matrix. **Results.** Networks with per-channel pooling and a linear classifier exhibit substantial intervention-size dependence despite numerical logit additivity. In a deeper architecture, additive reconstruction instead predicts a positive contrast when the observed contrast is negative ($\mathbf{+0.042}$ versus $\mathbf{-0.138}$). The disagreement depends on downstream architecture across 100 renderer blocks. The spatially permuted filter bank changes patching contrasts even though its rank, singular values, and Gram matrix are preserved; individual changes closely reproduce the architecture-averaged spatial difference. **Conclusion.** Intervention-size dependence alone does not demonstrate interactions among patched channels. Conversely, an additive approximation can reverse the inferred sign of a training-regime contrast. Distinguishing metric-induced effects from failures of logit additivity is essential when interpreting joint activation patches.

Keywords: Activation patching, mechanistic interpretability, convolutional neural networks, additivity, predefined filters

1 Introduction

An activation channel can have a small effect when patched alone and a different role when patched with other channels. Conversely, a joint patching curve can change sharply as channels are added even when their effects combine exactly in logit space. These possibilities create a practical problem: a change in a measured patching score does not, by itself, identify a change in how components interact. To interpret a joint intervention, we need to know what the individual interventions already predict.

Activation patching replaces an internal activation with its value from another input and measures the resulting output change. It is widely used to investigate the functional roles of model components, but its conclusions depend on the source and target inputs, the patched components, and the evaluation metric (Heimersheim and Nanda 2024; Zhang and Nanda 2024; Miller et al. 2024). The difficulty extends beyond choosing a metric. Zhang and Nanda (2024) compare simultaneous multi-layer patches with sums of individual-layer effects and find discrepancies. Vaidyanathan et al. (2026) derive interaction terms in causal mediation and multi-component patching, showing why effects measured separately need not predict a circuit's joint behavior. These interactions are already established. We ask when they are needed to explain a particular patching result.

We study that question in small CNNs whose first layer begins with predefined edge, corner, and ring filters. This setting lets us control the initial filters and the channels we patch. Structured spatial filters have a long history in wavelet representations, Gabor initialization, and constrained convolutional layers (Bruna and Mallat 2013; Özbulak and Ekenel 2018; Molaei and Shiri Ahmad Abadi 2020; Wang and Alkhalifah 2024; Linse et al. 2023; Orozco-Solis et al. 2024). These filters have recognizable spatial patterns, but a recognizable kernel does not establish how its activation channel contributes to a prediction. Weight similarity, representation of a concept, and influence on the output are different properties (Kim et al. 2018; Koh et al. 2020; Margeloiu et al. 2021; Nicolson et al. 2025; Sharma and Le 2026).

Our experiments compare two training regimes. A constant penalty retains the first-layer filters near their predefined bank; annealing the penalty releases them during training. Matched counterfactual images change a class-defining factor while keeping rendering conditions such as angle, scale, and noise fixed. We patch complete post-ReLU first-layer feature maps, vary the number of replaced channels, and evaluate how closely the patched output approaches the counterfactual output. We measure how the release-minus-retention difference changes between smaller and larger interventions. We sum the centered-logit changes from individual channel patches to predict each joint output, then evaluate that output with the same metric.

The comparison gives two different outcomes. In the smallest networks, individual changes reconstruct the joint output to numerical precision, yet the intervention-size contrast remains substantial. In a deeper network, the additive reconstruction predicts a positive contrast while the observed contrast is negative. Thus, the score can depend on intervention size even when joint logit changes equal the sum of individual changes. When the joint logit change differs from this sum, the discrepancy can reverse the sign of the contrast. A separate control permutes filter pixels while preserving pairwise filter inner products and singular values. Individual effects closely reproduce the contrast

averaged across architectures. Most of this average spatial difference can therefore be explained without a large departure from logit additivity.

We combine three types of evidence to distinguish these explanations in structured CNNs. Architectures with exact logit additivity show what can happen when individual changes add up. A comparison of 16 architectures shows where that prediction fails. A spatial control that preserves the Gram matrix tests whether matched filter inner products imply the same patching contrast. Together, these experiments determine when the sum of individual logit effects explains the reported joint patching contrast and when relying on that sum changes its interpretation.

2 Related work

2.1 Structured filters and functional interpretation

Wavelet scattering builds fixed multiscale representations (Bruna and Mallat 2013). Gabor-based CNNs initialize filters with Gabor kernels, maintain their structure during training, or learn their parameters (Özbulak and Ekenel 2018; Molaei and Shiri Ahmad Abadi 2020; Wang and Alkhalifah 2024; Orozco-Solis et al. 2024). Predefined spatial filters and frozen random convolutions also provide useful features for downstream learning (Linse et al. 2023; Gaudio et al. 2023; Gavrikov and Keuper 2023). Our study uses predefined filters to make first-layer structure controllable and inspectable. It evaluates the effect of intervening on the resulting channels rather than proposing another filter parameterization.

Concept-based explanations make a similar distinction between representing a concept and using it to produce an output. TCAV and concept bottlenecks connect model representations to specified concepts (Kim et al. 2018; Koh et al. 2020), while subsequent work examines the limits of concept alignment and its relationship to influence (Margeloiu et al. 2021; Nicolson et al. 2025; Sharma and Le 2026). Here, similarity to an edge, corner, or ring template describes the filter weights. The patching experiment measures a separate property of the trained network.

2.2 Patching metrics and joint interventions

Zhang and Nanda (2024) show that corruption methods and evaluation metrics can alter localization results. Their sliding-window analysis explicitly compares joint multi-layer patching with sums of single-layer effects. Heimersheim and Nanda (2024) discuss the interpretation of patching evidence and the different behavior of logit- and probability-based metrics. Miller et al. (2024) document sensitivity of circuit faithfulness to ablation methodology. These studies motivate checking both the intervention and its measurement before assigning a mechanistic interpretation.

Our comparison adds individual *logit changes*, rather than individual scalar patching scores. This distinction matters because squared reconstruction error and softmax can transform an additive output change into a non-additive score. By reconstructing the logits first, we can apply the same metric to the predicted and observed joint outputs.

2.3 Interaction effects and the scope of our baseline

Vaidyanathan et al. (2026) distinguish effects associated with individual mediators and their bypasses from cross-component interactions in jointly patched sets. Their analysis connects these terms to errors in ranking components and variation in transformer circuit faithfulness. It also gives a second-order approximation when the downstream function is smooth. Their decomposition directly informs comparisons of individual and joint interventions.

We reconstruct the centered-logit vector by measuring every individual channel change against the same base run and summing those changes. The residual is the difference between this prediction and the observed joint output. This residual does not estimate the single-mediator interaction term defined in that work, isolate effects along specific causal paths, or separate pairwise from higher-order interactions. Individual channel effects can themselves depend on the activations left unpatched. We compare this prediction with observed outputs in trained CNNs. The experiments include architectures with exact additivity when complete channels are replaced, a grid of 16 architectures, and a pixel-permuted filter bank. The reconstruction uses the actual output changes from channel replacements, so it does not require a local smoothness approximation for the ReLU network.

3 Method

3.1 Controlled inputs and training regimes

We use two 32×32 grayscale procedural classification tasks. In `single_shape`, the four classes are line, circle, triangle, and square. In `two_concepts`, circle and triangle presence define two binary factors and four classes. Rendering varies angle, line width, scale, translation, Gaussian noise, and occasional occlusion. A counterfactual pair changes a class-defining factor while keeping the other rendering conditions fixed. We focus on `two_concepts` and use `single_shape` to check whether similar patterns occur with a second renderer.

Each renderer block contains 128 training, 64 validation, and 64 test contexts sampled independently. A context specifies rendering conditions such as angle, scale, and noise. Rendering all four classes in every context gives 512, 256, and 256 images in the corresponding splits. Statistical analysis treats each block as one independent observation after averaging its initialization replicates.

All models begin with a bias-free $1 \to 16$ convolution with 9×9 kernels, same padding, and ReLU. The predefined bank contains eight oriented edge filters, four two-ray corners, and four rings. Each filter is centered and normalized to unit norm; the flattened 16×81 bank has numerical rank 10. Figure 1 shows the inputs, filters, and intervention location.

Training minimizes cross-entropy plus a normalized first-layer penalty,

$$\lambda(e)\,\frac{\|W - W_{\mathrm{ref}}\|_F^2}{\|W_{\mathrm{ref}}\|_F^2}, \tag{1}$$

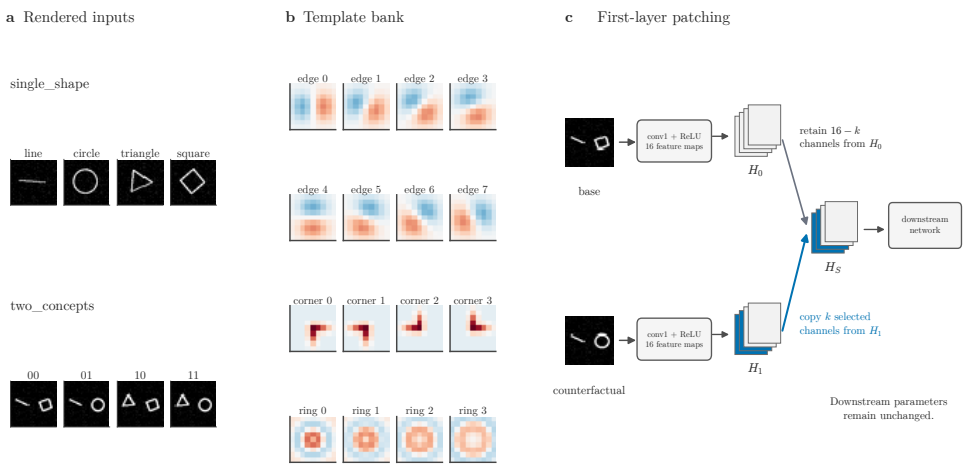


Fig. 1 A controlled setting for individual and joint patches. The renderer supplies matched counterfactual images; the first convolution starts from 16 edge, corner, and ring filters. Patching replaces selected complete post-ReLU feature maps in a base run with the corresponding maps from the counterfactual run. Individual and joint patches use the same trained network and input pair.

where W_{ref} is the initialization bank and e is the epoch. Retention keeps $\lambda = 1$ throughout 200 epochs. Release keeps $\lambda = 1$ through epoch 10, decreases it linearly to zero by epoch 80, and then trains without the penalty. Both regimes train the entire network with Adam, learning rate 0.003, and batch size 128. Downstream parameters can therefore change along with the first layer.

Figure 2 illustrates how the two schedules change the learned filters. This visual comparison establishes neither the causal role of a filter nor the additivity of its activation effects; those questions are assessed by the subsequent patching experiments.

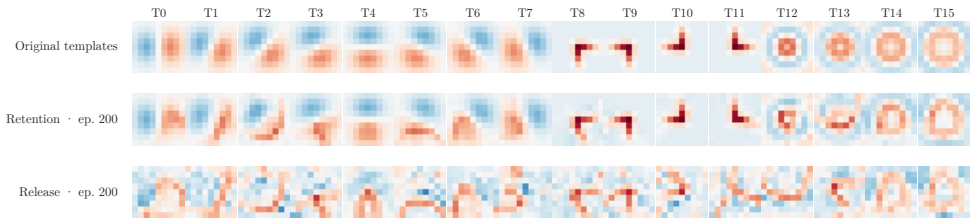


Fig. 2 Illustrative first-layer filters for `two_concepts`/TinyCNN, renderer block 2000. The same 16 original templates are shown above the retention and release weights at epoch 200. Learned kernels are aligned to templates by maximum-sum one-to-one assignment of centered, unit-norm signed cosine similarity; columns follow assigned template index. Each displayed kernel is centered and normalized to unit spatial norm, with one symmetric color scale across all rows. This example visualizes filter structure, not the functional effect of activation patching. The complete conditions and epochs are shown in Supplementary Fig. S5.

Filter–template similarity is signed cosine similarity after centering each spatial kernel and normalizing it to unit norm. For normalized learned filter $\bar{w}_j$ and template $\bar{t}_m$, the nearest-template score is

$$S_{\text{temp}}(W, T) = \frac{1}{16} \sum_{j=1}^{16} \max_m \bar{w}_j^\top \bar{t}_m. \tag{2}$$

A centered kernel with norm below 10^{-12} is treated as degenerate. The supplementary filter-similarity summaries also report scores obtained by matching each learned filter to a different template.

3.2 Architectures and spatial control

The grid of 16 architectures varies total convolutional depth (1, 2, or 4), downstream width (16 or 64), global max versus global average pooling (GMP/GAP), residual connections, and downstream BatchNorm. Tiny models pool immediately after the first convolution. Plain-2 adds one 3×3 convolution; Plain-4 adds three. Width-64 variants expand the first downstream convolution from 16 to 64 channels. Residual-4 has one skip from the first-layer activation to the third downstream convolution before its final ReLU. BN-2 and BN-4 are width-16 plain models with BatchNorm after each downstream convolution. All variants end in global pooling and a four-class linear classifier. Complete layer definitions and parameter counts are in Supplementary Section S1, *Architecture and training specification*.

The spatial control applies one common random permutation to the 81 coordinates of every filter. Writing the bank as $T \in \mathbb{R}^{16 \times 81}$ and the permutation matrix as Q, the permuted bank satisfies

$$(TQ)(TQ)^\top = TT^\top. \tag{3}$$

It also preserves row norms, rank, singular values, and the coefficients in each filter, including how often each value occurs, while changing their two-dimensional arrangement. For each bank, we train networks under both retention and release, allowing downstream parameters to adapt. We therefore compare trained networks rather than replacing filters in a fixed network.

3.3 Patching and output measures

For each matched pair, we cache the post-ReLU first-layer activations. We rank channels on the validation set by the absolute standardized difference between their global-max activations in the two counterfactual states. This ranking is used for every architecture, including GAP models. Test data are not used for selection. The original analyses patch the top k channels for $k \in \{1, 2, 4, 8\}$, with random same-size channel sets as controls. The reconstruction evaluates $k = 0, 1, \ldots, 16$.

Let $\ell \in \mathbb{R}^C$ be the class logits, with $C = 4$, and define centered logits

$$z = P\ell, \qquad P = I - \frac{\mathbf{1}\mathbf{1}^\top}{C}. \tag{4}$$

Centering subtracts the mean logit from every class logit and leaves softmax probabilities unchanged. Subscripts 0, 1, and S denote the base, counterfactual donor, and run patched on channel set S. For either centered logits or class probabilities, $q \in \{z, p\}$ with $p = \mathrm{softmax}(\ell)$, define

$$E_q(a, b) = \frac{1}{N} \sum_{i=1}^{N} \|q_a^{(i)} - q_b^{(i)}\|_2^2, \tag{5}$$

where i indexes evaluated pairs. Fidelity is

$$F_q(S) = 1 - \frac{E_q(S, 1)}{E_q(0, 1)}. \tag{6}$$

We denote centered-logit fidelity by F_{clogit} and the probability fidelity used in the initial analysis by F_{prob}. Fidelity is undefined when the sum of squared base–donor differences for the corresponding metric is below 10^{-12}. Values are not clipped to $[0, 1]$: a negative value means the patched output is farther from the donor than the base output is.

To check whether normalization drives the comparison, we also report unnormalized probability-error reduction,

$$R_{\mathrm{prob}}(S) = E_{\mathrm{prob}}(0, 1) - E_{\mathrm{prob}}(S, 1). \tag{7}$$

Thus F_{clogit} measures relative recovery of the donor's centered logits, while R_{prob} measures an absolute reduction in probability-space error.

For a metric M, write $\Delta^M(k) = M_{\mathrm{release}}(k) - M_{\mathrm{retention}}(k)$. The intervention-size contrast is

$$B^M = \frac{\Delta^M(4) + \Delta^M(8)}{2} - \frac{\Delta^M(1) + \Delta^M(2)}{2}. \tag{8}$$

Positive B means that the release-minus-retention difference is larger on average at $k = 4, 8$ than at $k = 1, 2$. It does not require release to outperform retention at every size, nor does it imply a monotone patching curve. Opposite signs of observed and reconstructed B give opposite conclusions about how this difference changes with intervention size. They need not imply opposite conclusions about classification accuracy or an individual channel's effect.

3.4 Predicting joint outputs from individual patches

For a fixed trained model and input pair, the single-channel change is measured against the same base run. Define

$$v = z_1 - z_0, \qquad d_j = z_{\{j\}} - z_0, \qquad a_S = \sum_{j \in S} d_j, \tag{9}$$

and reconstruct the joint output as

$$\widehat{z}_S = z_0 + a_S, \qquad r_S = z_S - \widehat{z}_S. \tag{10}$$

Observed metrics use z_S; reconstructed metrics use $\widehat{z}_S$. For probability metrics, we apply softmax *after* summing the logit changes. We never sum single-channel fidelities or probability-error reductions to predict a joint metric.

Even when $r_S = 0$, the joint fidelity need not equal the sum of single-channel fidelities. The additive centered-logit error is

$$\|v - a_S\|_2^2 = \|v\|_2^2 - 2v^\top a_S + \|a_S\|_2^2. \tag{11}$$

Both the magnitude of the summed change and its alignment with the base-to-donor change matter. Expanding $\|a_S\|_2^2$ produces cross-products between individual logit changes even in an exactly additive network. These cross-products arise from the squared-error metric, even when the network's logit changes are additive.

We summarize cancellation by

$$K(S) = 1 - \frac{\|a_S\|_2}{\sum_{j\in S}\|d_j\|_2}, \tag{12}$$

reported as undefined when the denominator is at most 10^{-12}. A larger $K(S)$ indicates more cancellation, but does not determine whether the summed change moves the output toward the donor. Residual energy is

$$\rho(S) = \frac{\sum_i \|r_S^{(i)}\|_2^2}{\sum_i \|v^{(i)}\|_2^2}. \tag{13}$$

It measures how far joint logit changes depart from the sum of single-channel changes, relative to the donor–base change. It is not a proportion of explained variance and can exceed one.

Tiny architectures provide a control with exact logit additivity. If h_j is channel j, ϕ_j is its independent spatial pooling operation, and A_j is its classifier column, their logits have the form $\ell(h) = b + \sum_j A_j\phi_j(h_j)$. Replacing complete channels in S changes this output by

$$\ell_S - \ell_0 = \sum_{j\in S} A_j\big[\phi_j(h_{1,j}) - \phi_j(h_{0,j})\big] = \sum_{j\in S}(\ell_{\{j\}} - \ell_0). \tag{14}$$

Centering is linear, so the same identity holds for z. Global-max pooling can be nonlinear within a channel and still satisfy this identity when complete channels are replaced. Exact additivity follows directly from the architecture, which makes these models a known control.

3.5 Experimental sequence and inference

Table 1 records the experimental sequence. The initial size test used 20 held-out renderer blocks. The architecture study and spatial-control study each used fresh samples of 100 blocks, with four initialization replicates averaged within block. The analysis of individual channel patches is post-hoc: it re-evaluates the saved models and blocks

Table 1 Experimental evidence and its role. The reconstruction uses existing checkpoints and renderer blocks.

Study	Blocks	Role
Intervention size	20	Held-out test of the release–retention size contrast in TinyCNN.
Architecture	100	Compare trained systems across 16 downstream architectures.
Spatial control	100	Compare original and pixel-permuted banks across four architectures.
Reconstruction	Reused	Compare predictions from individual patches with joint patches on saved models.

from these studies, rather than providing an independent replication. The contrast B always uses $k \in \{1, 2, 4, 8\}$, including when the reconstructed curves cover all sizes.

For the primary spatial comparison, we report 95% paired Student-t intervals across renderer blocks. We also test equivalence using pre-specified paired two one-sided tests (TOST) at $\alpha = 0.05$, equivalent to checking 90% intervals. We fixed symmetric equivalence margins before inspecting the new spatial-control results. Each margin is 20% of the mean absolute B in the earlier architecture study, using the same four architectures on `two_concepts`. This gives $\delta_{\text{clogit}} = 0.01704$ and $\delta_{R_{\text{prob}}} = 0.06594$. These are study-specific comparison thresholds, not externally validated measures of scientific importance.

For additivity summaries, we average initialization replicates within block and use 10,000 bootstrap resamples of renderer blocks. The paired comparisons between observed and reconstructed contrasts use Student-t intervals where stated. The spatial figure retains its original renderer-block intervals; these need not coincide numerically with the paired intervals used in the text.

4 Results

4.1 Intervention-size dependence under exact logit additivity

Retention keeps filters close to the templates, but filter similarity alone does not predict their patching effects. On `two_concepts`, TinyCNN's mean template similarity is 0.944 under retention and 0.574 under release; test accuracies are 98.05% and 99.45%. In the corresponding two-layer network, similarities are 0.999 and 0.896, with accuracies of 99.11% and 99.26%. The regularizer therefore changes how closely the filters match the templates, while both regimes classify the procedural images accurately.

In the held-out 20-block TinyCNN test, the release-minus-retention probability-fidelity differences at $k = 1, 2, 4, 8$ are -0.3459, -0.3317, -0.0221, and $+0.0065$. The size contrast is $B = +0.33098$, with a 95% interval of $[0.27368, 0.38827]$. Centered-logit fidelity and unnormalized probability-error reduction also give positive contrasts, $+0.17888$ and $+0.89441$, respectively. The comparison between training regimes therefore depends on which intervention size is evaluated.

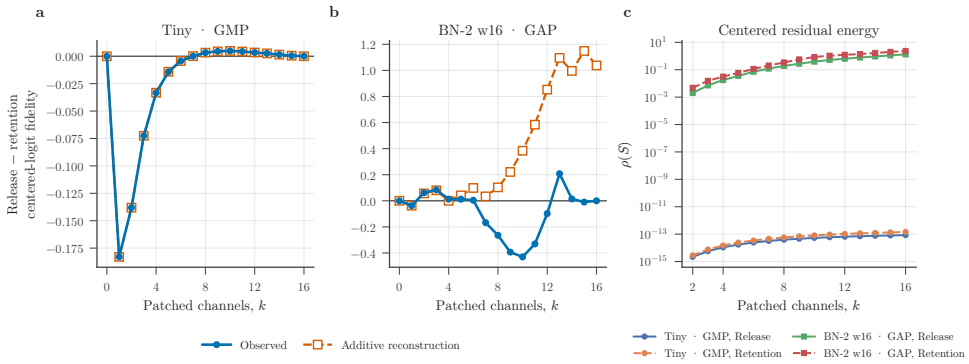


Fig. 3 The same reconstruction succeeds in one architecture and fails in another. (a) Tiny-GMP joint centered-logit fidelity contrasts are reconstructed to numerical precision from individual channel changes. (b) In BN2-GAP, observed and reconstructed release-minus-retention curves separate, yielding opposite signs of B. (c) Residual energy stays at numerical precision in Tiny and becomes substantial in BN2. All panels use the primary `two_concepts` task. Curves are renderer-block means without uncertainty bands; the paired BN2 difference and its interval are reported in the text. Reconstruction covers $k = 0, \ldots, 16$; B uses only $k = 1, 2, 4, 8$.

Tiny-GMP shows how this size dependence can occur under exact logit additivity. In the later 100-block Tiny-GMP analysis on `two_concepts`, observed and reconstructed B are both $+0.14558$ for centered-logit fidelity and both $+0.83151$ for probability-error reduction. The full curves agree to numerical precision (Fig. 3a). At $k = 16$, residual energy is only 8.41×10^{-14} under release and 1.45×10^{-13} under retention; across sizes it remains at numerical precision. The no-op patch reproduces the base logits, and the full patch reproduces the donor logits, both with zero error. These values come from the reused 100-block study, not the earlier 20-block confirmatory test.

Cancellation increases as more channels are patched: in the release regime, $K(S)$ rises from 0.054 at $k = 4$ to 0.130 at $k = 8$ and 0.193 at $k = 16$. This describes a reduction in the magnitude of the sum relative to individual magnitudes. Whether that reduction improves fidelity also depends on the alignment term in Eq. 11. Cancellation alone does not explain the curve. The directions and magnitudes of individual logit changes, together with the evaluation metric, can make the score depend strongly on intervention size even when the logit changes add up to numerical precision.

4.2 A joint effect can reverse the additive prediction

Individual logit changes do not always predict the joint response. In BN2-GAP on `two_concepts`, the observed centered-logit contrast is $B_{\text{obs}} = -0.13817$, whereas the reconstruction from individual logit changes predicts $B_{\text{add}} = +0.04177$ (Fig. 3b). Their paired difference across the same 100 renderer blocks is -0.17994, with a 95% Student-t interval of $[-0.20082, -0.15906]$. At $k = 8$, residual energy is 0.190 under release and 0.339 under retention. The reconstruction consequently gives the wrong sign for how the release–retention comparison changes between smaller and larger patches.

The reconstruction also fails on the secondary `single_shape` task: BN2-GAP gives -0.27012 observed versus $+0.34955$ reconstructed, and BN4-GAP gives $+0.37725$ versus $+1.64910$. These discrepancies cannot arise from applying a nonlinear metric to identical logits: observed and reconstructed outputs already differ before metric evaluation. For these channel replacements, joint logit changes differ from the sum of single-channel changes. This result does not identify which downstream operation causes the difference.

The architecture grid puts these examples in context. Across 16 architectures and 100 independent renderer blocks, repeated-measures tests show differences in B across architectures. For centered-logit fidelity, $F(15, 1485) = 77.09$, with partial $\eta^2 = 0.438$. For probability-error reduction, $F(15, 1485) = 656.70$, with partial $\eta^2 = 0.869$. Architecture differences remain under Greenhouse–Geisser correction, within-block permutation tests, Friedman tests, and random channel orders. Figure 4 shows the original size curves and the architecture-level contrasts; Fig. 6a compares those contrasts with predictions from individual channel patches.

Under GMP, adding the first downstream convolution accounts for much of the change from Tiny to plain models. Adding further convolutions has a smaller effect. GAP gives a different pattern. Residual connections produce comparatively small changes. The effects of width and BatchNorm depend on the other architectural choices. The grid does not establish a universal effect of depth or show that BatchNorm alone causes the residual.

4.3 Filter banks with identical Gram matrices give different patching contrasts

The pixel-permuted control tests whether preserving the bank's rank and pairwise filter inner products is enough to reproduce the original patching contrast. For channels selected on the validation set in `two_concepts`, centered-logit B is $+0.05743$ higher for the structured bank than for the permuted bank, averaged across four architectures. Its 95% paired Student-t interval is $[+0.05116, +0.06369]$, and its 90% interval $[+0.05219, +0.06266]$ lies outside the pre-specified equivalence margin ± 0.01704. Identical bank Gram matrices therefore do not yield equivalent centered-logit contrasts under this criterion.

The probability-error result meets the equivalence criterion. The mean difference is $+0.04563$, with a 95% paired Student-t interval of $[+0.03728, +0.05397]$. Its 90% interval, $[+0.03864, +0.05261]$, lies within the wider margin ± 0.06594. The 95% interval excludes zero, while the 90% interval satisfies the study-specific equivalence criterion.

The metrics have different scales and equivalence margins. These results do not establish that the spatial effect is generally important or negligible.

The direction of the spatial effect also depends on architecture. For channels selected on the validation set, the difference is positive in Tiny-GMP, Tiny-GAP, and Plain-2-GAP, but negative in Plain-2-GMP. Random subsets give smaller differences averaged across architectures (Fig. 5). The control therefore identifies sensitivity to the original arrangement in these training regimes, rather than a general advantage of visually recognizable filters.

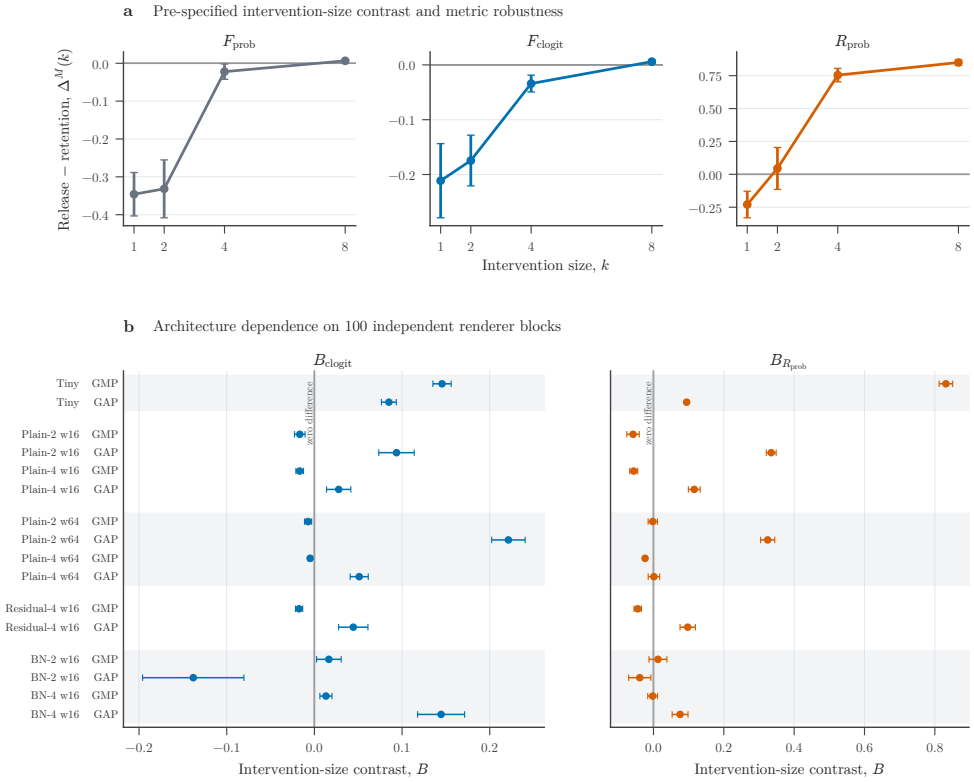


Fig. 4 A patching contrast depends on both intervention size and the trained architecture. Top: release-minus-retention curves in the 20-block `two_concepts`/TinyCNN study, under probability fidelity, centered-logit fidelity, and probability-error reduction; bars are 95% intervals across renderer blocks. Bottom: architecture-specific B in the fresh 100-block `two_concepts` study; points average four initialization replicates within block before aggregation, and bars are 95% intervals. Positive B denotes a larger release-minus-retention difference at $k = 4, 8$ than at $k = 1, 2$, not a uniform advantage for release.

4.4 Individual effects largely reproduce the average spatial difference

Does the spatial contrast require a large departure from logit additivity? For centered-logit B averaged across architectures, the observed difference between the structured and permuted banks is +0.05743. The reconstruction gives +0.05710. The paired observed-minus-reconstructed difference is +0.00033, with a 95% Student-t interval of $[-0.00074, +0.00140]$. Individual logit effects, passed through the same fidelity metric, closely reproduce the average contrast (Fig. 6b). The interval contains zero, but this alone does not establish equivalence.

For probability-error reduction, the contrasts are +0.04563 observed and +0.04105 reconstructed. Their paired difference is +0.00458, with a 95% interval of $[+0.00202, +0.00714]$. Across individual architectures, differences between observed and reconstructed spatial contrasts range from −0.00473 to +0.00604 for centered

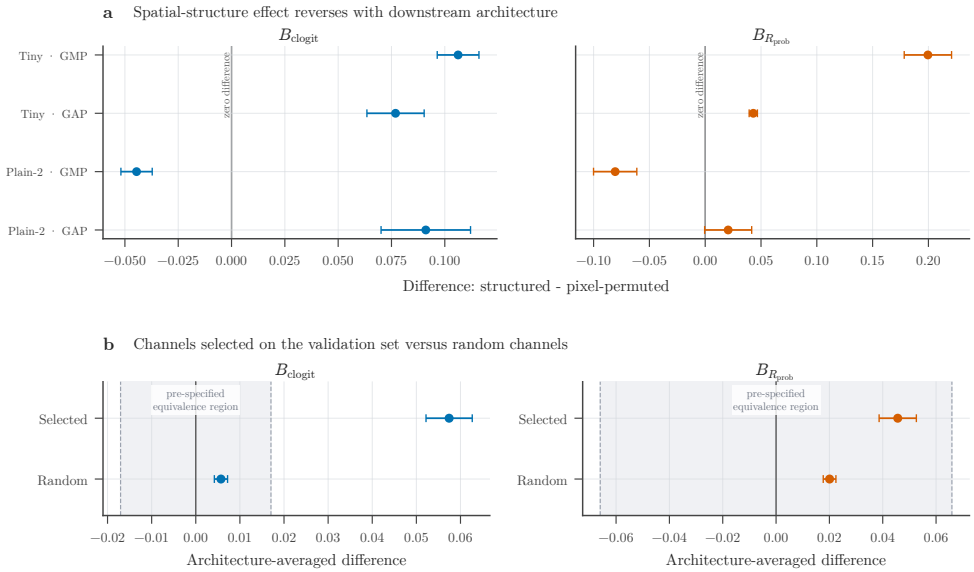


Fig. 5 Preserved filter inner products, different patching contrasts. The common spatial permutation preserves the bank's Gram matrix, rank, singular values, row norms, and the coefficients in each filter, including how often each value occurs. (a) Structured-minus-permuted B differs in sign across architectures; bars are 95% intervals across renderer blocks. (b) Differences averaged across architectures for selected and random channel sets, with 90% TOST intervals. Gray bands show the study-specific equivalence margins, ± 0.01704 for centered-logit fidelity and ± 0.06594 for probability-error reduction. All comparisons use `two_concepts` and 100 renderer blocks.

logits and from -0.00635 to $+0.02466$ for probability error; both Tiny variants are essentially at zero. The average agreement therefore does not imply agreement in every architecture. It shows that summing individual logit changes explains much of this particular spatial comparison.

5 Discussion

To interpret a joint patch, we must distinguish changes in the *score* from departures from additivity in the *joint logit response*. A squared-error fidelity can vary strongly with patch size because individual changes point in different directions and because the metric contains cross-products. Probability metrics additionally transform the summed logits through softmax. These transformations remain when logit additivity is exact. The Tiny control shows that size dependence alone is insufficient evidence of channel interaction in that architecture.

The BN2-GAP result demonstrates the opposite failure: assuming that individual effects suffice can reverse the size contrast. The reconstruction error changes the sign of the reported comparison between training regimes. Residual energy alone, however, does not determine whether a joint patch improves fidelity. Since $z_S = z_0 + a_S + r_S$, the donor error is $\|v - a_S - r_S\|_2^2$. Both the residual's magnitude and its direction relative to $v - a_S$ matter. A nonzero residual shows that the individual logit changes do

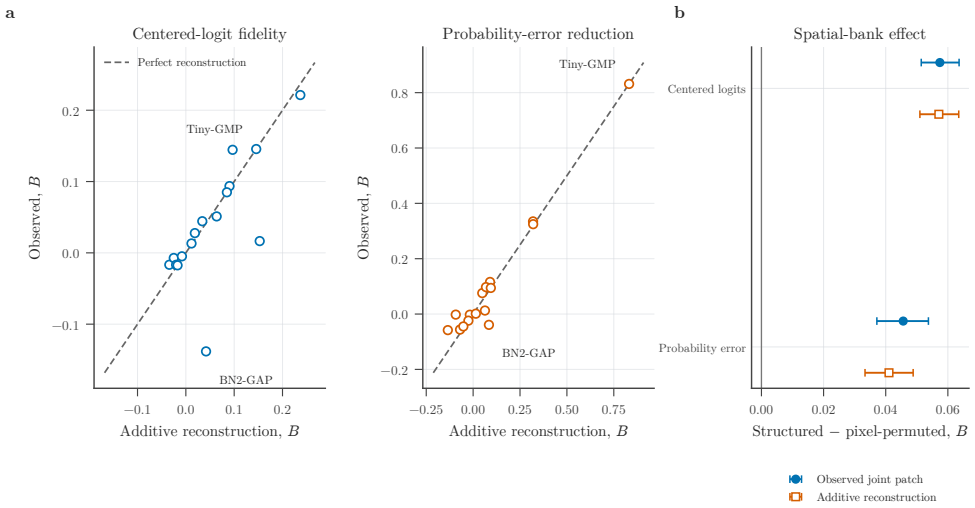


Fig. 6 Where predictions from individual patches agree with joint patches. (a) Observed versus reconstructed B across 16 architectures on `two_concepts`; the diagonal denotes agreement. Highlighted Tiny-GMP and BN2-GAP cases correspond to Fig. 3. (b) Structured-minus-permuted B, averaged across the four spatial-control architectures. Horizontal bars show 95% bootstrap intervals for each estimate, resampling renderer blocks. The text reports paired differences between observed and reconstructed contrasts with Student-t intervals. Overlap of the separate intervals in the figure does not test the paired difference.

not add up for the evaluated patches. It does not show that the interaction improves fidelity or identify a unique mechanism.

This interpretation is consistent with the interaction analysis of Vaidyanathan et al. (2026): separately measured components can fail to predict collective behavior. Our comparison includes trained networks where individual patches reconstruct the joint logits exactly, and others where the reconstruction error changes the sign of the reported contrast. A zero residual for the evaluated channel sets does not establish context-independent causal importance, nor does it exclude higher-order interactions whose contributions cancel. The reconstruction checks additivity for the specified intervention. Claims about specific mechanisms may still require tests of other channel combinations or causal paths.

The spatial control extends the same reasoning to filter structure. The original and permuted banks have the same pairwise filter inner products. Their two-dimensional arrangements differ, as do the responses learned from them. Their patching contrasts can differ even when individual logit changes closely predict the joint outputs on average. Filter similarity alone therefore does not determine the effect of an individual channel patch or how those effects combine in the downstream network. The experiment establishes this distinction for the bank and permutation studied; it does not rank spatially structured filters as universally more interpretable.

In practice, reconstruct the joint logits from individual interventions, apply the original metric, and compare the reconstructed contrast with the observed contrast using a paired analysis. This comparison shows whether the sum of individual logit

changes, evaluated with the same metric, predicts the joint response. The interpretation still depends on intervention size, how donor and base inputs are constructed, how channels are selected, and the downstream architecture.

5.1 Scope and limitations

The tasks are synthetic, the networks are small, and the spatial control uses one predefined bank and one common permutation. These choices support controlled comparisons, but do not establish the frequency or magnitude of the phenomena in large models trained on natural images. Retention and release train the entire network differently, allowing downstream parameters to adapt along with the first layer. We do not match classification accuracy across architectures. The comparisons describe the trained networks, rather than isolating an architecture effect at fixed accuracy.

We do not compare against three state-of-the-art methods on standard classification benchmarks because this study does not propose a new predictor or training algorithm whose predictive performance is being benchmarked. Instead, it asks whether the summed logit changes from individual channel patches predict joint interventions. Comparisons of classification accuracy with other methods would not directly distinguish effects of the patching metric from failures of logit additivity. We therefore evaluate two controlled procedural tasks, architectures with analytically additive logits, a 16-architecture grid, and a pixel-permuted filter-bank control. These experiments test the specified mechanistic question; they do not establish generalization to state-of-the-art models or natural-image datasets.

The largest reconstruction discrepancies in this grid include BatchNorm architectures, but they cannot be assigned specifically to BatchNorm. With stored statistics in evaluation mode, BatchNorm is affine; non-additivity can arise from its composition with channel mixing, ReLU, pooling, and later layers. A layer-specific causal explanation would require further interventions.

Matched donors are valid rendered images, but the patched hybrid state may still fall outside ordinary activation distributions. Selection uses global-max first-layer activations even in GAP models; random-channel analyses reduce dependence on this ranking without removing it. Finally, the reconstruction is post-hoc and uses the original saved blocks. Its paired comparison checks additivity in those existing results; it is not a new confirmatory replication. These boundaries define the claims that the present evidence supports.

6 Conclusion

Individual activation patches can be sufficient to explain a joint patching contrast, or insufficient even to predict its sign. In controlled CNNs, an intervention-size contrast persists under exact logit additivity, while a deeper architecture produces a residual that reverses the additive prediction. Permuting filter pixels while preserving the Gram matrix also changes the patching contrast. Individual effects closely reproduce the centered-logit difference averaged across architectures.

The practical implication is: a size-dependent score does not establish a channel interaction, and an additive prediction should be checked before it is used to interpret a

joint intervention. Comparing observed joint outputs with predictions from individual patches distinguishes these two cases. It also gives a more specific account of the studied results than filter similarity or patching magnitude alone.

Supplementary information. A standalone Supplementary Information PDF provides the architecture/training specification and supporting numerical summaries. The accompanying reproducibility artifact contains the frozen protocols, analysis code, and block-level or aggregate data underlying every reported main-text figure and statistic. Raw training checkpoints and per-model evaluator files are retained separately because of their size.

Declarations

Funding Not applicable.

Competing interests The author declares no competing interests.

Ethics approval and consent to participate Not applicable. The study uses procedurally generated images and does not involve human participants or animals.

Consent for publication Not applicable.

Data availability Experimental images are procedurally generated using the renderer documented in the research code. Archived aggregate and block-level results, including the inputs needed to regenerate all six manuscript and five supplementary figures, are publicly available in the versioned reproducibility artifact at doi:10.5281/zenodo.23432463. The artifact enables selected numerical checks but does not independently rerun every inferential analysis. Original training checkpoints and per-model evaluator outputs are not included.

Code availability Selected training, evaluation, and analysis code, the frozen experimental protocols, figure generators, and the standalone reproduction runner are archived in the versioned Zenodo release doi:10.5281/zenodo.23432463 (Espino Montelongo 2026). From these archived inputs, the runner regenerates eleven figure PDFs and independently checks selected reported statistics; full retraining requires additional raw experiment outputs and checkpoints.

Author contributions H.E.M. conceived the study, developed the methodology and software, conducted the experiments and analyses, prepared the visualizations, and wrote and revised the manuscript.

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